CSE 200: Online-1 on LaTeX (C1)

Time: **50 Minutes**
Total Marks: **50**

Instructions

- Reproduce the following article using LaTeX exactly as it is.
- You do not need to reproduce this instruction page.
- Ensure that all text formatting, table of contents, lists, nested lists, tables, equations, matrices, figures, pseudocode, citations, and references are implemented as presented in the article using appropriate LaTeX commands.
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Mark Distribution

Topic	**Marks**
Document Structure, Title, Table of Contents & Two-column Layout	4
Text Formatting, Nested Lists, Colors & Emphasis	5
Equations, Alignments & Matrices	12
Colored Table with Multirow Formatting	7
Two-column Figure, Rotation, Caption & Label	4
Pseudocode and Complexity Expression	5
Theorem, Proof & Mathematical Environment	4
Cross-references	3
Footnote & Hyperlink	2
Citations & Bibliography	2
Overall Presentation & Successful Compilation	2
Total	**50**

The Riemann Hypothesis

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Contents

1 A Function Built from Integers

1.1 The Riemann Zeta Function

For a complex number $s = \sigma + it$ with $\sigma > 1$, the Riemann zeta function is

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \text{ prime}} \left(1 - \frac{1}{p^s}\right)^{-1}. \tag{1}$$

The sum–product identity is the EULER PRODUCT and links $\zeta(s)$ with primes [1]. See the Clay Mathematics Institute.[1]

1.2 A Classical Special Value

At $s = 2$, Equation (1) gives the classical identity

$$\zeta(2) = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots = \frac{\pi^2}{6},$$
$$\prod_{p \in \{2,3,5,7\}} \left(1 - \frac{1}{p^2}\right)^{-1} \approx 1.595 < \frac{\pi^2}{6} \approx 1.645.$$

More prime factors improve the approximation.

1.3 Two Mathematical Worlds

Figure 1 connects zeta zeros with prime distribution.

2 Zeros, Symmetry, and the Critical Line

2.1 Trivial and Nontrivial Zeros

The `trivial zeros` are

$$-2, -4, -6, \ldots$$

and the `non-trivial zeros` lie in the `critical strip`

$$0 < \mathrm{Re}(s) < 1.$$

(A) **Trivial zeros**

- ∤ They occur at negative even integers.
- ∤ Their locations are completely understood.

(B) **Nontrivial zeros**

- ∋ They lie inside the critical strip.
- ∋ They occur in symmetric families.
- ∋ Their symmetries generate the following companion points:
  - (i) ρ;
  - (ii) $1 - \rho$;
  - (iii) $\overline{\rho}$;
  - (iv) $1 - \overline{\rho}$.
- ∋ Their horizontal locations are central to the Riemann Hypothesis.

[1] The Riemann Hypothesis is a Millennium Prize Problem.

2.2 The Functional Equation

A symmetric form of the zeta function is

$$\xi(s) = \frac{1}{2} s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s), \qquad \xi(s) = \xi(1-s). \tag{2}$$

Equation (2) reflects zeros across $\sigma = 1/2$:

$$\begin{bmatrix} \sigma' \\ t' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sigma \\ t \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

The linear part of the reflection is the matrix

$$R = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad R^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2. \tag{3}$$

Hence R is an involution.

2.3 The First Few Zeros

The line $\mathrm{Re}(s) = \frac{1}{2}$ is the *critical line*.

Table 1: The first four positive ordinates of nontrivial zeros.

<table>
<tr><th>Index</th><th>γ_n</th><th>Real Part</th><th>Imaginary Part</th></tr>
<tr><td>1</td><td>14.134725</td><td rowspan="4">$\frac{1}{2}$</td><td>14.134725</td></tr>
<tr><td>2</td><td>21.022040</td><td>21.022040</td></tr>
<tr><td>3</td><td>25.010858</td><td>25.010858</td></tr>
<tr><td>4</td><td>30.424876</td><td>30.424876</td></tr>
</table>

The first two zeros form the matrix

$$Z_2 = \begin{bmatrix} \frac{1}{2} & 14.134725 \\ \frac{1}{2} & 21.022040 \end{bmatrix}, \tag{4}$$

Each row stores one zero.

2.4 A Numerical Inspection Procedure

Algorithm 1 tests candidate real parts.

Algorithm 1 Check candidate zeros against the critical line

Require: Candidate zeros $Z = (\rho_1, \rho_2, \ldots, \rho_m)$ and tolerance $\varepsilon > 0$
Ensure: TRUE if every candidate satisfies $\left|\mathrm{Re}(\rho_i) - \frac{1}{2}\right| \le \varepsilon$

```
allOnLine ← TRUE
for i ← 1 to m do
    if |Re(ρ_i) − 1/2| > ε then
        allOnLine ← FALSE
        break
    end if
end for
return allOnLine
```

Its running time is

$$T(m) = \Theta(m). \tag{5}$$

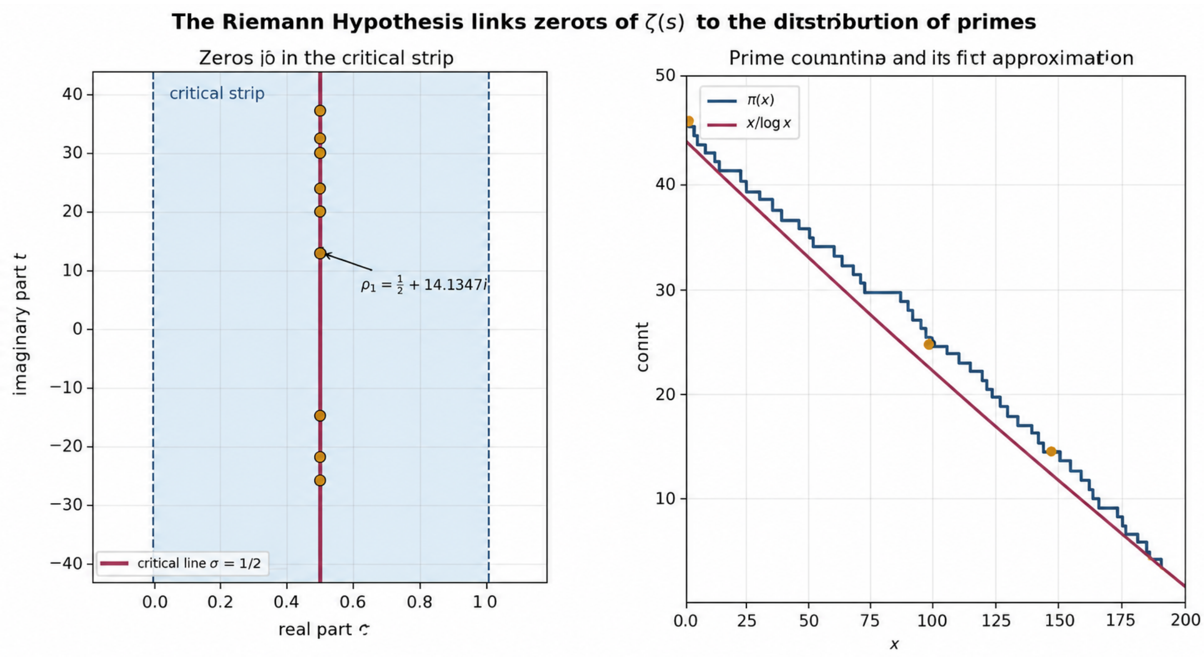

Figure 1: The critical strip and the relation between prime counting and analytic approximation.

3 The Riemann Hypothesis

3.1 Formal Statement

The Riemann Hypothesis states

$$\boxed{\zeta(\rho) = 0, \quad 0 < \mathrm{Re}(\rho) < 1 \quad \Longrightarrow \quad \mathrm{Re}(\rho) = \frac{1}{2}.} \tag{6}$$

Equation (6) remains unproved [2].

3.2 Logical Interpretation

It asserts necessity, not sufficiency.

Thus,

$$\zeta(\rho) = 0 \quad \Longrightarrow \quad \mathrm{Re}(\rho) = \frac{1}{2}$$

for nontrivial zeros; the converse fails in general.

3.3 An Equivalent Error Statement

Define the Chebyshev function

$$\psi(x) = \sum_{p^k \le x} \log p.$$

Theorem 3.1 (Prime-distribution error bound)**.** *The Riemann Hypothesis is equivalent to*

$$\psi(x) = x + O\left(\sqrt{x}\log^2 x\right). \tag{7}$$

Proof. The explicit formula expresses the error through the nontrivial zeros. The stated bound is equivalent to all such zeros having real part $\frac{1}{2}$. □

Theorem 3.1 links zeros and prime-counting error.

4 Why Prime Numbers Care

Primes appear irregularly, but zeta zeros control that irregularity. *RH sharply limits the prime-counting error.*

4.1 Mathematical Formulation

Let $\pi(x) = \#\{p \le x : p \text{ is prime}\}$ denote the prime-counting function. The prime number theorem states that

$$\pi(x) \sim \frac{x}{\log x}, \qquad \lim_{x\to\infty} \frac{\pi(x)\log x}{x} = 1. \tag{8}$$

A sharper approximation is

$$\mathrm{Li}(x) = \int_2^x \frac{dt}{\log t}. \tag{9}$$

Under RH,

$$\pi(x) = \mathrm{Li}(x) + O\left(\sqrt{x}\log x\right). \tag{10}$$

Hence zeta zeros govern prime-counting fluctuations.

References

[1] B. Riemann, "On the number of primes less than a given magnitude," *Monatsberichte der Berliner Akademie*, 1859. English translation of the original German paper.

[2] H. M. Edwards, *Riemann's Zeta Function*. Academic Press, 1974.